Class 1: Difference-in-Differences

In-class student worksheet

Name: _____ Date: _____

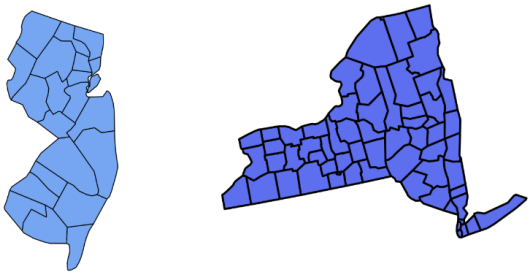
Learning objectives:

By the end of today's class, we will be able to:

1. Quantify if more generous UI lead workers to collect benefits for longer?
2. Translate policy reforms into testable causal questions.
3. Compare correlation vs. causality.
4. Understand how to apply difference-in-differences.
5. Group activity: DiD application with Stata or R

1. Why is a simple comparison insufficient?

Suppose State A has higher average UI benefits and longer average UI receipt than State B.



	NJ	NY
Avg. UI benefit ($)	628.29	553.97
Avg. UI duration (weeks)	19.3	17.9

Can we conclude that higher benefits caused longer UI receipt? Give **two other explanations** for the difference between the states.

2. A policy change: New Jersey and Pennsylvania

In 1996, New Jersey temporarily made 13 additional weeks of UI benefits available. Pennsylvania did not introduce the same extension. We compare regular-benefit exhaustion rates before and after the extension.

Outcome: the percentage of claimants who use up their *regular* UI benefits. It is not the unemployment rate and does not measure exhaustion of the additional extension.

A. Identify the design. On the axes below, plot both states' before/after outcomes. Add NJ's counterfactual after-period point. Label the estimated policy effect.



Treated state: _____ Comparison state: _____

What is the treatment? _____

What is the outcome of their interest? _____

B. Complete the table. All differences are in **percentage points**.

	Before	After	After − Before
New Jersey	49.9%	49.4%	
Pennsylvania	31.8%	29.0%	
NJ − PA			

C. Calculate the DiD estimate using both approaches.

$$\widehat{\tau} = (\text{NJ after} - \text{NJ before}) - (\text{PA after} - \text{PA before}) = _____$$

D. Interpret your estimate in one sentence.

3. When is the comparison credible?

1. Parallel trends: _____

    - How do we know there are parallel trends?

2. The policy changes is the only important change that happened in NJ is this time period.

    - Example: If NJ experienced a larger economic downturn than PA at the same time as the extension, would the DiD estimate be biased upward or downward?

4. DiD with Linear Regression:

Regression connection. In the equation below, circle the coefficient that corresponds to your DiD estimate:

$$Y_{i,t} = \alpha + \beta NJ_i + \lambda After_t + \tau(NJ_i \times After_t) + \varepsilon_{i,t}.$$

$NJ_i = 1$ for NJ and 0 for PA; $After_t = 1$ after the extension and 0 before.

Table 1: Predicted Regular UI Exhaustion Rates

	Before	After	After − Before
NJ			
PA			
NJ − PA			